

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Marine productivity & fishing - Set 1

Mark scheme

AQA spec 3.5.2.1-3.5.2.2; Biological resources; textbook pp. 338-351

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• In winter nutrients are high (storms disturb sediments, deep mixing) but light is too low for photosynthesis • In spring light rises while nutrients are still high so growth is rapid - the spring bloom (peak about May) • The bloom uses up nutrients (falling to 10 by July); dead plankton sink out of the photic zone carrying nutrients to the seabed • So nutrients become limiting in summer despite high light; a smaller autumn bloom follows when storms return nutrients while light is still adequate	4
01.2	• Warm surface water is stable (no seasonal mixing / storms) so nutrients lost to deep water in sinking dead organisms are not replaced • Nutrient (especially phosphate) concentrations in the photic zone remain very low - the limiting factor	2
01.3	• Upwelling (offshore winds, seamounts, currents deflected upwards) brings cold nutrient-rich deep water into the photic zone • Coastal waters receive nutrients washed off the land in rivers and storms disturb shallow sediments • Nutrients plus light in the shallow photic zone allow high algal productivity supporting rich food webs and fisheries (eg Peru anchovy)	3
01.4	• Phosphates have low solubility and no atmospheric reservoir; they are only added by river runoff or upwelling and are lost to sediments	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 338-339; AQA spec 3.5.2.1.	

Question	Marking guidance	Marks
02.1	• Demersal trawl: seabed species - cod, haddock, plaice, shrimps, scampi; chains / weights damage the seabed (mix aerobic and anaerobic layers, destroy sea fans, corals, seagrass); high by-catch in mixed fisheries; high fuel use • Purse seine: shoaling pelagic species - tuna, sardines, herring, anchovy; can remove whole shoals; dolphins associated with tuna shoals may be caught • Pelagic long-line: tuna, squid; albatrosses (long-lived, low breeding rate) drown on baited hooks; sharks and turtles caught	6
02.2	• Shrimp trawls use very small mesh so crabs, molluscs, sea urchins, starfish, juvenile fish cannot escape - by-catch can be 99% of the catch • Drift nets are non-selective curtains near the surface catching whales, dolphins, turtles, sharks	2
02.3	• Lost or discarded nets and traps continue to catch and kill organisms; dead animals act as bait attracting more • Biodegradable ropes / panels so lost traps fall apart; radio transmitters on traps so they can be recovered; net recovery schemes	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 340-343; AQA spec 3.5.2.2.	
03.1	• k-selected: very slow growth, does not breed until 30, few eggs, so the population cannot replace losses • Fish reach catchable size (20) before they breed (30) so fishing can remove the entire future breeding population • Deep-water aggregations easily located and caught by trawling • Recovery takes decades so fisheries have had to be closed	4
03.2	• Herring breed at 2 years - the same age they become catchable - so there is always a breeding population and many young • Cod are catchable at 2 but do not mature until 4, so heavy fishing removes most fish before they breed	2
03.3	• Larger numbers of smaller / younger fish are caught, or bigger boats, nets and sonar increase catch effort • Masks the decline of the breeding population until a sudden collapse	2
03.4	• Neither is a target species but both are caught in demersal trawls; skate now extinct in the Baltic and critically endangered elsewhere • Very late maturity (skate 11 years, Greenland shark 100-150 years, living over 400) and few young mean even low by-catch mortality exceeds replacement	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 341-342, 348-349; AQA spec 3.5.2.2.	
04.1	• $600 + (90 + 150) - (220 + 80) = 540$ • Stock fell by 60 so the catch exceeds the MSY / unsustainable	2
04.2	• $A + G - M = 90 + 150 - 80 = 160$	1
04.3	• Catch is landed and weighed at ports so it is recorded • Other terms need data on fish in the sea: populations are mobile (crossing national boundaries), very unevenly distributed • Representative sampling of a large enough proportion of the area is impractical / expensive; survival of eggs and larvae varies greatly between years	3

Question	Marking guidance	Marks
04.4	• CPUE: if more effort (boat-days, engine power, net size) is needed to land the same catch, the population is declining even though total catch is maintained; a better measure than catch size which technology can prop up • Falling mean size / age (from scale or otolith growth rings) shows older adults are being removed faster than replaced and the catch depends on small young fish	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 349-351; AQA spec 3.5.2.2.	
05.1	• Normal diagonal mesh closes as the net fills / stretches so small fish cannot escape; square-mesh panels do not close so juveniles escape to grow and breed; TEDs let turtles out; larger mesh sizes generally • Minimum size lets fish reach breeding age before capture (eg UK lobster minimum carapace above breeding size, cod 35 cm); maximum size / V-notching egg-carrying females protects large, highly fecund breeding adults (Hebridean lobsters over 145 mm, Nassau grouper) • NTZs (Lundy, Lamlash Bay, Chagos, Great Barrier Reef) protect breeding populations and habitats; surplus young spill over and increase surrounding catches; damaged communities recover • Closed seasons, ideally in the breeding season, let fish grow and breed; fishermen may oppose because shoaling breeders are easy to catch	8
05.2	• Pelagic species form single-species shoals so fishing stops when the quota is reached • In mixed fisheries many species are caught together; once one species' quota is reached it must be discarded (dead) while fishing for others continues	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 344-348; AQA spec 3.5.2.2.	
06.1	<i>Any 3 from:</i> • Sinkers / weighted lines so hooks sink quickly below the reach of albatrosses • Setting lines at night when birds are not feeding • Bird-scaring streamers / decoys • Curved hooks that catch tuna but not birds	3
06.2	• Effort: limits on boat size, engine power, days at sea, number of licences; decommissioning payments to reduce fleet size where overfishing is unavoidable • Methods: bans or length limits on drift nets; bans on demersal trawling over sensitive seabed (Rockall Bank, Hatton Bank deep-water reefs) • Shorter trawl tows so turtles / by-catch survive (over 99% survive under 10 minutes)	3
06.3	• Eggs / larvae are reared in captivity through the planktonic stage when most die (predation, failure to find habitat) • Juveniles are released when survival chances are good, increasing recruitment to the adult population	2
06.4	• Egg / larval survival varies greatly between years so adult numbers do not predict recruitment • Several poor years would lower the MSY; egg surveys allow catches to be reduced in advance rather than after a sudden collapse	2

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 344-348; AQA spec 3.5.2.2.	
07.1	- By-catch = $5.9 + 1.9 = 7.8$ t; total = 8.4 t $7.8 / 8.4 \times 100 = 92.9\%$ (accept 93%)	2
07.2	The nets have a very small mesh size so few non-target animals can escape	1
07.3	- They breed at an old age (skate about 11 years; Greenland shark 100-150 years) and produce few young - They may reach a catchable size before they are sexually mature, so fishing can remove the whole breeding population before it has bred - Populations recover slowly, unlike r-selected species (herring, mackerel) which breed young and produce many young; they are caught as by-catch in demersal trawls so fishing continues even though they have no commercial value	3
07.4	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) - Impacts: population decline of k-selected species, by-catch (immature target fish, over-quota fish, non-commercial species; drift nets - whales, dolphins, turtles; longlines - albatrosses; pair trawls - porpoises), ghost fishing, seabed damage by demersal trawls, coral and seagrass damage, dynamite fishing, food-web effects - Catch quotas: limit weight landed; work well for single-species pelagic shoals but in mixed fisheries over-quota fish are discarded dead so the quota fails to protect them - Gear design: mesh size and square-mesh escape panels let small fish escape (diagonal mesh closes under drag); turtle exclusion devices; acoustic pingers for dolphins; curved hooks, decoys, sinkers and night setting to protect albatrosses; biodegradable / radio-tracked traps to reduce ghost fishing - Restricted effort: limits on boat size, engine power and days at sea; decommissioning; restricted methods - drift-net bans, demersal trawling bans on sensitive seabeds (Rockall Bank, Hatton Bank) - Evaluation: enforcement at sea is difficult; some methods reduce catch or raise costs; quotas need reliable stock data; effectiveness depends on species (single vs mixed shoals) and on international cooperation as fish migrate; as local stocks decline fishing moves to more distant waters - Justified conclusion weighing effectiveness of different methods	9
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 341-345; AQA spec 3.5.2.2.	